

Junseo Lee – Curriculum Vitae

CONTACT INFORMATION

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EDUCATION

Harvard University, Kenneth C. Griffin Graduate School of Arts and Sciences (GSAS) Cambridge, MA
Ph.D. in Quantum Science and Engineering Sep. 2026 – Present

- Advisors: [Anurag Anshu](#) and [Sitan Chen](#) (Computer Science); [Jordan Cotler](#) (Physics)
- Affiliations: [Harvard Quantum Initiative](#); [Theory of Computation Group](#)
- Supported by the Kwanjeong Doctoral Study Abroad Fellowship

Yonsei University Seoul, Korea
B.S. in Electrical and Electronic Engineering Mar. 2019 – Feb. 2023

- Thesis: *Combinatorial designs for information theory* (Advisor: Hong-Yeop Song)
- Supported by the Hyundai Motor CMK Science and Technology Scholarship

Chungnam Science High School Gongju, Korea
Concentration in Mathematics; Early Honors Graduate Mar. 2017 – Dec. 2018

RESEARCH POSITIONS

NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI), MIT Cambridge, MA
Junior Investigator (Nominator/Advisor: Jordan Cotler) May. 2026 – Present

Seoul National University (SNU) Seoul, Korea
Research Fellow, *Institute of Computer Technology* (Host: Kabgyun Jeong) Apr. 2026 – Aug. 2026
Research Affiliate, *Research Institute of Mathematics* (Host: Kabgyun Jeong) Jan. 2023 – Mar. 2026
Undergraduate Researcher, *Research Institute of Mathematics* (Advisor: Kabgyun Jeong) Mar. 2020 – Dec. 2022

Republic of Korea Army Seoul, Korea
Research Scientist (Technical Research Personnel), *Mandatory Military Service* Mar. 2023 – Mar. 2026

PUBLICATIONS ([Google Scholar Profile](#))

*Equal contribution. †Authors listed alphabetically.

- (19) Francisco Escudero Gutiérrez, **Junseo Lee**[†], and Sebastian Zur, “Hamiltonian locality testing and certification do not achieve the Heisenberg limit,” (2026). [arXiv:2610.03205](#).
- (18) **Junseo Lee**[†], and Chirag Wadhwa, “Optimal purity estimation with incoherent measurements,” (2026). [arXiv:arXiv:2609.40248](#).
- (17) Myeongjin Shin, **Junseo Lee**, Iman Marvian, and Yu Tong, “Emergent prethermal symmetries for scalable Hamiltonian learning,” (2026). [arXiv:2609.27181](#).
- (16) Myeongjin Shin, **Junseo Lee**, and Changhun Oh, “Heisenberg-limited Hamiltonian learning without short-time control,” [PRX Quantum](#) (2026). doi: 10.1103/kjvf-jgqj.
– Long talk at the [26th Asian Quantum Information Science Conference \(AQIS 2026\)](#).
- (15) Andreas Bluhm, Matthias C. Caro, Francisco Escudero Gutiérrez, **Junseo Lee**[†], Aadil Oufkir, Cambyse Rouzé, and Myeongjin Shin, “Certifying and learning local quantum Hamiltonians,” (2026). [arXiv:2603.29809](#).

- Extends and subsumes my earlier work [arXiv:2512.09778](#).
 - Contributed talk at the [21st Conference on the Theory of Quantum Computation, Communication and Cryptography \(TQC 2026\)](#).
- (14) **Junseo Lee**[†] and Myeongjin Shin, “Optimal certification of constant-local Hamiltonians,” (2025). [arXiv:2512.09778](#).
 - (13) Marco Fanizza, Vishnu Iyer, **Junseo Lee**[†], Antonio Anna Mele, and Francesco Anna Mele, “Efficient learning of bosonic Gaussian unitaries,” (2025). [arXiv:2510.05531](#).
 - Contributed talk at the [29th Annual Conference on Quantum Information Processing \(QIP 2026\)](#).
 - (12) Donghwa Ji, **Junseo Lee**, Myeongjin Shin, IlKwon Sohn, and Kabgyun Jeong, “Bounding quantum uncommon information with quantum neural estimators,” [Quantum Science and Technology](#) **11**, 015001 (2026). [DOI: 10.1088/2058-9565/ae18f4](#).
 - (11) Kartik Anand, Kabgyun Jeong, and **Junseo Lee**[†], “Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy,” (2025). [arXiv:2506.19792](#).
 - (10) Myeongjin Shin*, **Junseo Lee***, Seungwoo Lee, and Kabgyun Jeong, “Resource-efficient algorithm for estimating the trace of quantum state powers,” [Quantum](#) **9**, 1832 (2025). [DOI: 10.22331/q-2025-08-27-1832](#).
 - (9) Mingyu Lee, Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, “Mutual information maximizing quantum generative adversarial networks,” [Scientific Reports](#) **15**, 32835 (2025). [DOI: 10.1038/s41598-025-18476-y](#).
 - (8) Myeongjin Shin*, Seungwoo Lee*, **Junseo Lee***, Donghwa Ji, Hyeonjun Yeo, and Kabgyun Jeong, “Disentanglement provides a unified estimation for quantum entropies and distance measures,” [Physical Review A](#) **110**, 062418 (2024). [DOI: 10.1103/PhysRevA.110.062418](#).
 - (7) Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, “Estimating quantum mutual information through a quantum neural network,” [Quantum Information Processing](#) **23**, 57 (2024). [DOI: 10.1007/s11128-023-04253-1](#).
 - (6) **Junseo Lee** and Kabgyun Jeong, “Quantum Rényi entropy functionals for bosonic gaussian systems,” [Physics Letters A](#) **490**, 129183 (2023). [DOI: 10.1016/j.physleta.2023.129183](#)
 - (5) **Junseo Lee**, Hyeonjun Yeo, and Kabgyun Jeong, “Weighted p -Rényi entropy power inequality: Information theory to quantum Shannon theory,” [International Journal of Theoretical Physics](#) **62**, 253 (2023). [DOI: 10.1007/s10773-023-05512-8](#)
 - (4) **Junseo Lee** and Kabgyun Jeong, “High-dimensional private quantum channels and regular polytopes,” [Communications in Physics](#) **31**, 189 (2021). [DOI: 10.15625/0868-3166/15762](#)
 - (3) Kabgyun Jeong, **Junseo Lee**, *et al.*, “Single qubit private quantum channels and 3-dimensional regular polyhedra,” [New Physics: Sae Mulli](#) **68**, 232 (2018). [DOI: 10.3938/NPSM.68.232](#)
 - Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

- (2) **Junseo Lee**, “Assessing Quantum Integer Factorization Performance with Shor’s Algorithm,” In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*, Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa, [CRC Press](#) (2024). [DOI: 10.1201/9781003475286](#)

Patents

- (1) Kabgyun Jeong, Myeongjin Shin, and **Junseo Lee**, “Method for estimating quantum mutual information through a quantum neural network,” [Korea Patent Open No. 10-2026-0009068](#) (2024). [DOI: 10.8080/1020240091151](#)

SELECTED HONORS AND AWARDS

Funding and Fellowships

- Doctoral Study Abroad Fellowship, *Kwanjeong Foundation*, 2026–2030

- Graduate Fellowship, *Harvard University*, 2026
- Academic Travel Grant, *Hyundai Motor CMK Foundation*, 2022
- Science and Technology Scholarship (full tuition and stipend), *Hyundai Motor CMK Foundation*, 2021–2022

Academic Excellence

- High Honors, *Yonsei University*, 2022
- Honors, *Yonsei University*, 2020–2021
- Science Scholarship, *JH Foundation*, 2017
- Academic Excellence Scholarship, *Asan Future Scholarship Foundation*, 2016

Additional Honors and Awards

- Selected Paper Award, Finance and Economics Contest, *DB Cultural Foundation*, 2022
- Best Tutor Award, *Yonsei University*, 2021–2022
- Third Prize, Undergraduate Research Exhibition, *Korean Physical Society*, 2021
- Bronze Award, Humantech Paper Award, *Samsung Electronics*, 2018
- Best Translator Award, *NAVER Connect Foundation & Khan Academy*, 2018
- Pre-Undergraduate Research Fellowship, *Korea Advanced Institute of Science and Technology*, 2017–2018
- Honorable Mention (National) & Gold Award (Regional), *Korean Olympiad in Informatics (KOI)*, 2016

PROFESSIONAL ACTIVITIES

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Portfolio:** Nature Communications, npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik, Mathematical Methods in the Applied Sciences, Statistics and Probability Letters, Algorithmica

Conference Reviewer

- ACM-SIAM Symposium on Discrete Algorithms (SODA), 2027
- IEEE Symposium on Foundations of Computer Science (FOCS), 2026
- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Organizing

- Organizer, [Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems](#), 2026
- Co-organizer, [Quantum Information Theory Seminar](#), *Seoul National University*, 2024–2025
- Co-organizer, [Quantum AI Hackathon](#), *Kakao Enterprise & Jeonju University*, 2025

Committee Service

- Poster Session Committee Member, National Undergraduate Quantum Conference, *Seoul National University*, 2026
- Selection Committee Member, [Quantum Internship Program](#), *National Information Society Agency & Korea Quantum Industry Center*, 2024–2025

Outreach and Community Resources

- Creator and Maintainer, [Quantum Learning Theory Zoo](#), 2025–present
- Facilitator (Mentor), Korea Scholar's Conference for Youth (KSCY), *Yonsei University*, 2019

TEACHING *Teaching materials available at: harris-junseo-lee.github.io. †University–Industry Research Internship.

- Organizer and Instructor, [Quantum Learning Theory for Bosonic and Fermionic Systems](#), *Quantum Information Science Club Association*, Winter 2026*
- Instructor, [Quantum Complexity Reading Group](#), *Quantum Information Science Club Association*, Fall 2025*
- Instructor, [Quantum Learning and Complexity Theory](#), *Quantum Information Science Club Association*, Summer 2025*
- (External) Instructor†, AAA559: College of Informatics Internship (II) (Graduate Course), *Korea University*, Fall 2025
- (External) Instructor†, AAA558: College of Informatics Internship (I) (Graduate Course), *Korea University*, Fall 2025
- (External) Instructor†, SW4343: Software Field Placement (I), *Korea Aerospace University*, Fall 2024
- Teaching Assistant, YCS1009: Change the World through Programming, *Yonsei University*, Fall 2022
- Teaching Assistant, YCS1002: Software Programming, *Yonsei University*, Fall 2022
- Teaching Assistant, EEE1108: Engineering Information Processing, *Yonsei University*, Fall 2021
- Course Tutor, MAT2016: Engineering Mathematics (III), *Yonsei University*, Spring 2022 (Best Tutor Award)
- Course Tutor, MAT1012: Engineering Mathematics (II), *Yonsei University*, Fall 2021 (Best Tutor Award)

RESEARCH VISITS

- Harvard University (supported by Anurag Anshu and the Harvard Quantum Initiative), Mar. 2026

SELECTED TALKS

*Online talk.

Research Talks

“Heisenberg-limited Hamiltonian learning without short-time control”

- Invited talk, *Technische Universität Hamburg (Hamburg University of Technology)*, Jun. 2026*
- Long contributed talk, 26th Asian Quantum Information Science Conference (**AQIS 2026**), *Korea Advanced Institute of Science and Technology (KAIST)*, Aug. 2026

“Certifying and learning local quantum Hamiltonians”

- Invited talk, Quantum Meets (Online Seminar Series), Jun. 2026*
- Invited talk, 7th QNRC Joint Workshop, *Korea Institute of Science and Technology Information (KISTI)*, May 2026
- Contributed talk, 21st Conference on the Theory of Quantum Computation, Communication and Cryptography (**TQC 2026**), *Delta Hotels Sherbrooke Conference Centre (Québec, Canada)*, Sep. 2026

“Optimal certification of constant-local Hamiltonians”

- Invited talk, Quantum Software Lab Seminar, *University of Edinburgh*, Mar. 2026*
- Contributed talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026

“Efficient learning of bosonic Gaussian unitaries”

- Invited talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026
- Invited talk, N³etFraST Workshop, *Korea Institute of Science and Technology Information (KISTI)*, Nov. 2025
- Invited talk, Quantum Data Science & AI Lab Seminar, *Yonsei University*, Nov. 2025
- Contributed talk, 29th Annual Conference on Quantum Information Processing (**QIP 2026**), *University of Latvia*, Jan. 2026 (presented under the title “Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators”)

“Resource-efficient algorithm for estimating the trace of quantum state powers”

- Invited talk, *Electronics & Telecommunications Research Institute (ETRI)*, Dec. 2024
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Dec. 2024*

- Invited talk, IBM-Yonsei Qiskit Fall Fest, *Yonsei University*, Nov. 2024*
- Invited talk, KISTI-KU-SNU Joint Workshop, *Seoul Biohub*, Oct. 2024
- Contributed talk, Annual Meeting of the Korean Mathematical Society (KMS), *Sungkyunkwan University*, Oct. 2024
- Poster, 28th Annual Conference on Quantum Information Processing (QIP 2025), *Raleigh Convention Center (NC, United States)*, Feb. 2025

“Mutual information maximizing quantum generative adversarial network”

- Invited talk, Triangle Quantum Computing Seminar, *NC State University Quantum Initiative*, Nov. 2023*

“Estimating quantum mutual information through a quantum neural network”

- Invited talk, CS Katha Barta, *National Institute of Science Education and Research (Bhubaneswar, India)*, Aug. 2023*

“Quantum Rényi entropy functionals for bosonic Gaussian systems”

- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

“High-dimensional private quantum channels and regular polytopes”

- Invited talk, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Aug. 2021*
- Contributed talk, Winter Meeting of the Optical Society of Korea (OSK), *Daejeon Convention Center*, Feb. 2022
- Contributed talk, Fall Meeting of the Korean Physical Society (KPS), Oct. 2021*
- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

Tutorials and Public Lectures

“Getting Started in Quantum Information Science”

- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Apr. 2026
- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Aug. 2024

“Learning theory in ∞ -dimensional quantum systems”

- Invited talk, Team QST Summer Workshop, *Seoul National University*, Aug. 2025

“Introduction to quantum machine learning”

- Invited talk, Healthcare & Research Team Seminar, *Amazon Web Services (AWS Seoul, Korea)*, Mar. 2025

“Quantum machine learning models for drug library generation”

- Invited talk, Quantum Computing and Monte Carlo Workshop, *Yonsei University*, Aug. 2024

“QMA $\stackrel{?}{=}$ NP: The NLTS theorem and the quantum PCP conjecture”

- Invited talk, Center for Quantum Network’s Channel Capacity Summer Workshop, *Seoul National University*, Jul. 2024

“Minimal data may be sufficient for quantum artificial intelligence”

- Invited talk, Department of Mathematical Sciences Seminar, *Seoul National University*, Jun. 2023*